

17- Display your MAC address by 2 different ways.

```
sayedmohamed@sayedmohamed-Precision-3510: ~  
sayedmohamed@sayedmohamed-Precision-3510:~$ ip link show  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT  
group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
2: enp0s31f6: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state  
DOWN mode DEFAULT group default qlen 1000  
    link/ether 18:db:f2:48:4e:03 brd ff:ff:ff:ff:ff:ff  
3: wlp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mod  
e DORMANT group default qlen 1000  
    link/ether b8:8a:60:f6:c4:44 brd ff:ff:ff:ff:ff:ff  
sayedmohamed@sayedmohamed-Precision-3510:~$ ip link show eth0  
Device "eth0" does not exist.  
sayedmohamed@sayedmohamed-Precision-3510:~$ ip link show enp0s3  
Device "enp0s3" does not exist.  
sayedmohamed@sayedmohamed-Precision-3510:~$ ip link show wlp2s0  
3: wlp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mod  
e DORMANT group default qlen 1000  
    link/ether b8:8a:60:f6:c4:44 brd ff:ff:ff:ff:ff:ff  
sayedmohamed@sayedmohamed-Precision-3510:~$  
sayedmohamed@sayedmohamed-Precision-3510:~$
```

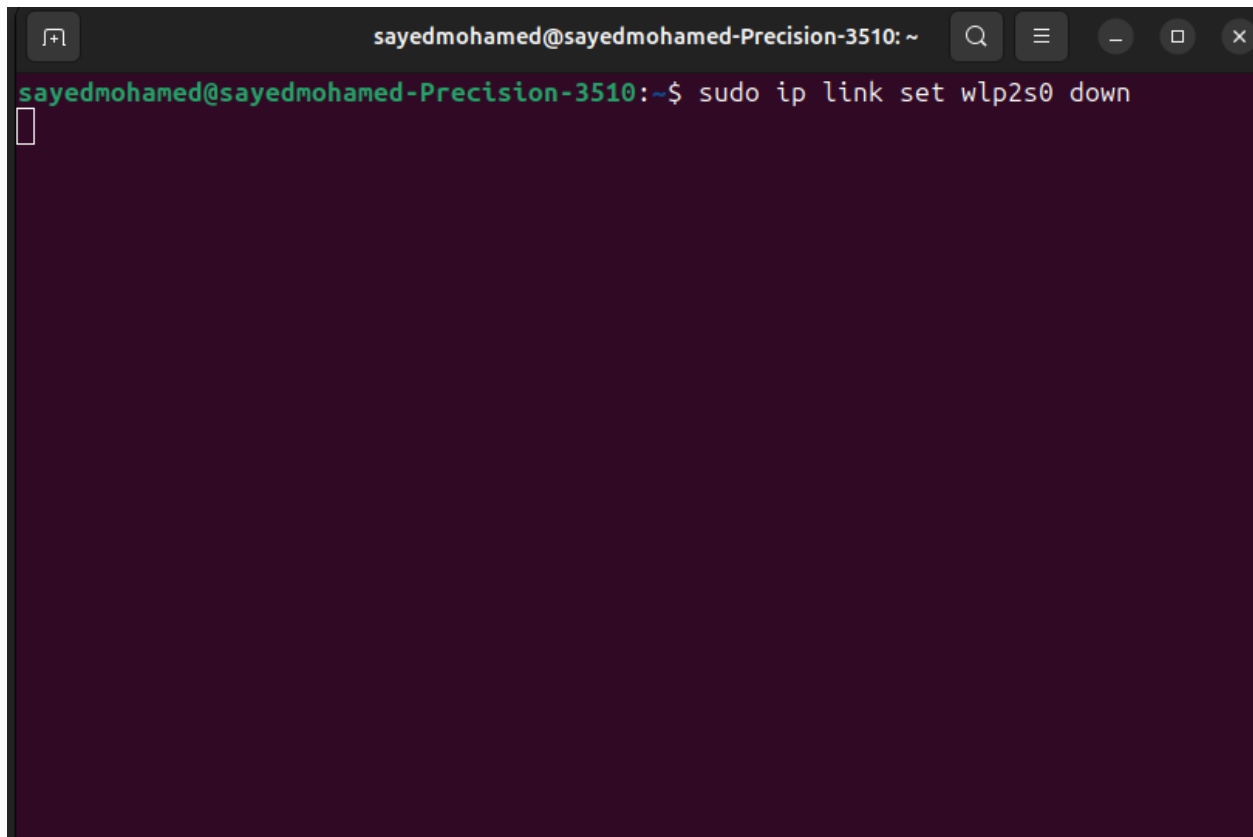
18- Display the network settings of all active interfaces.

```
sayedmohamed@sayedmohamed-Precision-3510: ~  
sayedmohamed@sayedmohamed-Precision-3510:~$ ip -brief addr show up  
lo UNKNOWN 127.0.0.1/8 ::1/128  
enp0s31f6 DOWN  
wlp2s0 UP 192.168.1.12/24 fdf4:e3fb:8ca0:7000:56a:fc3:14e  
4:44ba/64 fdf4:e3fb:8ca0:7000:79b5:472e:50b6:9a5a/64 fe80::e043:3792:179d:d24e/64  
sayedmohamed@sayedmohamed-Precision-3510:~$
```

19- Display the network setting of all interfaces both active inactive.

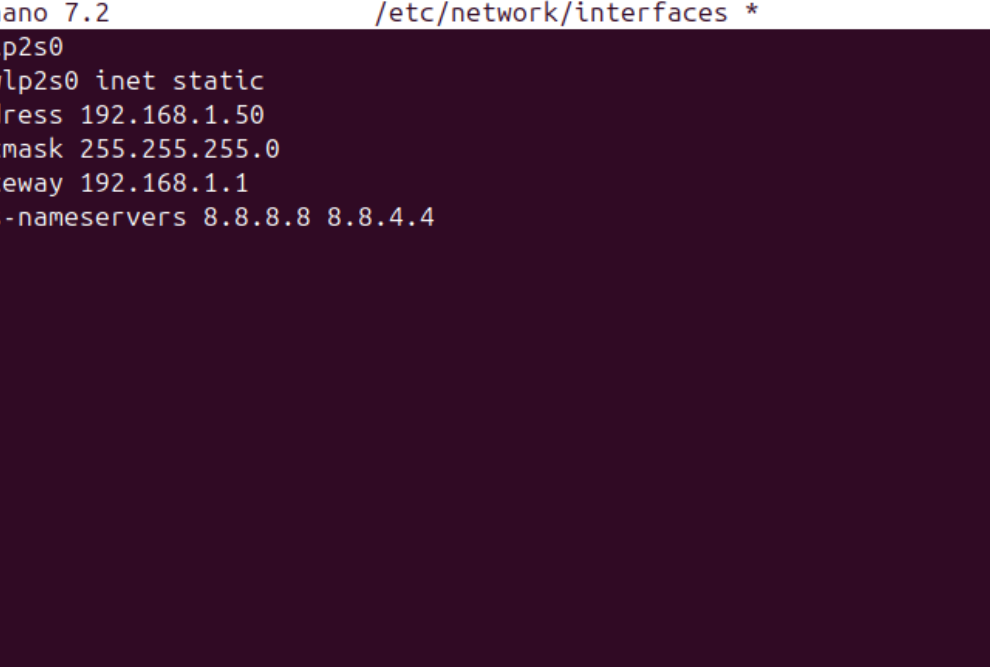
```
sayedmohamed@sayedmohamed-Precision-3510: ~  
sayedmohamed@sayedmohamed-Precision-3510:~$ ip addr show  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: enp0s31f6: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000  
    link/ether 18:db:f2:48:4e:03 brd ff:ff:ff:ff:ff:ff  
3: wlp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000  
    link/ether b8:8a:60:f6:c4:44 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.1.12/24 brd 192.168.1.255 scope global dynamic noprefixroute wlp2s0  
        valid_lft 82892sec preferred_lft 82892sec  
    inet6 fdf4:e3fb:8ca0:7000:56a:fc3:14e4:44ba/64 scope global temporary dynamic  
        valid_lft 6484sec preferred_lft 2884sec  
    inet6 fdf4:e3fb:8ca0:7000:79b5:472e:50b6:9a5a/64 scope global dynamic mngtmpaddr noprefixroute  
        valid_lft 6484sec preferred_lft 2884sec
```

20- Bring your interface down.

A terminal window with a dark background. The title bar at the top shows the user 'sayedmohamed' and the host 'sayedmohamed-Precision-3510'. The terminal displays a green prompt 'sayedmohamed@sayedmohamed-Precision-3510:~\$' followed by the command 'sudo ip link set wlp2s0 down' in white text. A white cursor is positioned at the end of the command line.

```
sayedmohamed@sayedmohamed-Precision-3510:~$ sudo ip link set wlp2s0 down
```

21- Configure your network card to have static IP.



```
GNU nano 7.2 /etc/network/interfaces *
auto wlp2s0
iface wlp2s0 inet static
    address 192.168.1.50
    netmask 255.255.255.0
    gateway 192.168.1.1
    dns-nameservers 8.8.8.8 8.8.4.4
```

22- Bring your interface up.

```
sayedmohamed@sayedmohamed-Precision-3510:~$ sudo ip link set wlp2s0 up
```

23- Verify your network setting using ifconfig command

```
sayedmohamed@sayedmohamed-Precision-3510: ~  
sayedmohamed@sayedmohamed-Precision-3510:~$ ifconfig  
enp0s31f6: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500  
    ether 18:db:f2:48:4e:03 txqueuelen 1000 (Ethernet)  
    RX packets 0  bytes 0 (0.0 B)  
    RX errors 0  dropped 0  overruns 0  frame 0  
    TX packets 0  bytes 0 (0.0 B)  
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0  
    device interrupt 16  memory 0xec300000-ec320000  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 1920  bytes 242664 (242.6 KB)  
    RX errors 0  dropped 0  overruns 0  frame 0  
    TX packets 1920  bytes 242664 (242.6 KB)  
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0  
  
wlp2s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
```

24- Configure your network card to have dynamic IP using network manager command.

```
sayedmohamed@sayedmohamed-Precision-3510: ~  
    inet6 fe80::e043:3792:179d:d24e prefixlen 64 scopeid 0x20<link>  
    inet6 fdf4:e3fb:8ca0:7000:56a:fc3:14e4:44ba prefixlen 64 scopeid 0x0<  
global>  
    inet6 fdf4:e3fb:8ca0:7000:79b5:472e:50b6:9a5a prefixlen 64 scopeid 0x0  
<global>  
    ether b8:8a:60:f6:c4:44 txqueuelen 1000 (Ethernet)  
    RX packets 76247 bytes 54073199 (54.0 MB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 84322 bytes 26099334 (26.0 MB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
sayedmohamed@sayedmohamed-Precision-3510:~$ nmcli con show  
NAME                                UUID                                TYPE      DEVICE  
WE_B34855                          0e56cddb-b4b9-4641-b770-0cce16edc748 wifi      wlp2s0  
lo                                  400d080f-bd6d-46d4-97d8-c052466bcfc4 loopback  lo  
A32 الخطأ ELsayed                   3b8c610e-288b-4cac-a524-bf951571b573 wifi      --  
Creativa_Zagazig                   6da24547-87c5-4e23-b806-419974707b43 wifi      --  
Hotspot                             a0b8167b-b34b-4974-a57a-59073724f30c wifi      --  
Redmi Note 11S                     c2bf4588-9a0f-4a02-962f-69877ccce9a0 wifi      --  
TEdataF3F3DD                       c8d93876-752b-4ef3-9277-77331cad3b67 wifi      --  
Wired connection 1                 d564a80f-9f03-38c1-9230-e615b46c5928 ethernet --  
sayedmohamed@sayedmohamed-Precision-3510:~$ sudo nmcli con mod "Wi-Fi connection  
1" ipv4.method auto #use DHCP
```

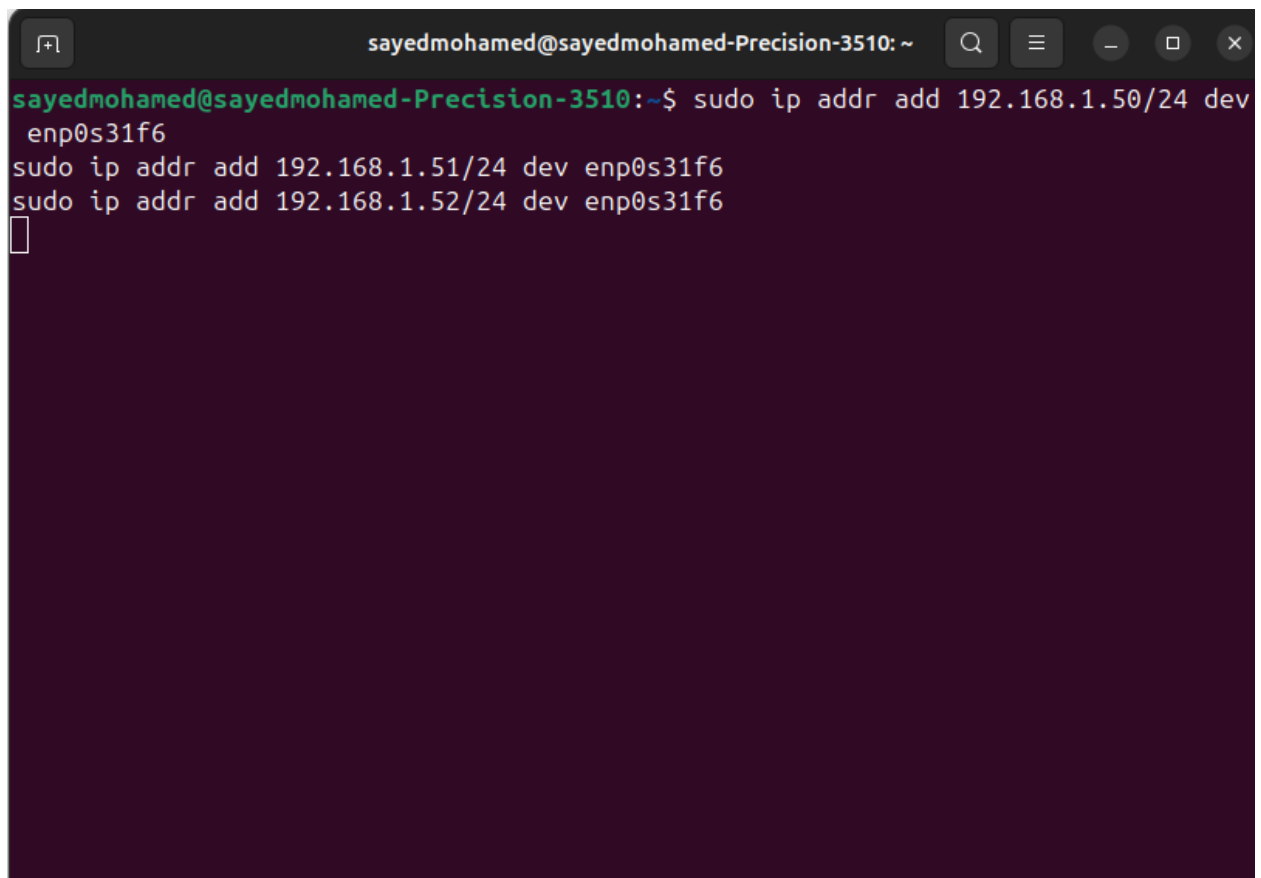
```
sayedmohamed@sayedmohamed-Precision-3510: ~  
    inet6 fe80::e043:3792:179d:d24e prefixlen 64 scopeid 0x20<link>  
    inet6 fdf4:e3fb:8ca0:7000:56a:fc3:14e4:44ba prefixlen 64 scopeid 0x0<  
global>  
    inet6 fdf4:e3fb:8ca0:7000:79b5:472e:50b6:9a5a prefixlen 64 scopeid 0x0  
<global>  
    ether b8:8a:60:f6:c4:44 txqueuelen 1000 (Ethernet)  
    RX packets 76247 bytes 54073199 (54.0 MB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 84322 bytes 26099334 (26.0 MB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
sayedmohamed@sayedmohamed-Precision-3510:~$ nmcli con show  
NAME                                UUID                                TYPE      DEVICE  
WE_B34855                          0e56cddb-b4b9-4641-b770-0cce16edc748 wifi      wlp2s0  
lo                                  400d080f-bd6d-46d4-97d8-c052466bcfc4 loopback  lo  
A32 الخطأ ELsayed                   3b8c610e-288b-4cac-a524-bf951571b573 wifi      --  
Creativa_Zagazig                   6da24547-87c5-4e23-b806-419974707b43 wifi      --  
Hotspot                             a0b8167b-b34b-4974-a57a-59073724f30c wifi      --  
Redmi Note 11S                      c2bf4588-9a0f-4a02-962f-69877ccce9a0 wifi      --  
TEdataF3F3DD                       c8d93876-752b-4ef3-9277-77331cad3b67 wifi      --  
Wired connection 1                  d564a80f-9f03-38c1-9230-e615b46c5928 ethernet --  
sayedmohamed@sayedmohamed-Precision-3510:~$ sudo nmcli con up "Wi-Fi connection  
1" #restarting the connection
```

25- Check using ifconfig then check its configuration file.


```
sayedmohamed@sayedmohamed-Precision-3510: ~  
sayedmohamed@sayedmohamed-Precision-3510:~$ ifconfig  
enp0s31f6: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500  
    ether 18:db:f2:48:4e:03 txqueuelen 1000 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
    device interrupt 16 memory 0xec300000-ec320000  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 1971 bytes 250529 (250.5 KB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 1971 bytes 250529 (250.5 KB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
wlp2s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.1.12 netmask 255.255.255.0 broadcast 192.168.1.255  
    inet6 fe80::e043:3792:179d:d24e prefixlen 64 scopeid 0x20<link>  
    inet6 fdf4:e3fb:8ca0:7000:56a:fc3:14e4:44ba prefixlen 64 scopeid 0x0<  
global>  
    inet6 fdf4:e3fb:8ca0:7000:78b5:472e:50b6:9a5a prefixlen 64 scopeid 0x0
```

26- Has a problem on ubuntu

27- Configure your network card to have 3 IPs and check that they are all working using ifconfig command.

A terminal window with a dark background and a title bar. The title bar contains the text 'sayedmohamed@sayedmohamed-Precision-3510: ~' and standard window control icons (search, menu, zoom, close). The terminal shows three lines of text: a green prompt 'sayedmohamed@sayedmohamed-Precision-3510:~\$' followed by the command 'sudo ip addr add 192.168.1.50/24 dev enp0s31f6', then 'sudo ip addr add 192.168.1.51/24 dev enp0s31f6', and finally 'sudo ip addr add 192.168.1.52/24 dev enp0s31f6'. A white cursor is positioned at the end of the third line.

```
sayedmohamed@sayedmohamed-Precision-3510: ~
sayedmohamed@sayedmohamed-Precision-3510:~$ sudo ip addr add 192.168.1.50/24 dev
enp0s31f6
sudo ip addr add 192.168.1.51/24 dev enp0s31f6
sudo ip addr add 192.168.1.52/24 dev enp0s31f6
█
```

28- Change your host name in your global network file.

```
sayedmohamed@sayedmohamed-Precision-3510: ~  
GNU nano 7.2 /etc/hostname  
sayedmohamed-Precision-3510  
  
[ Read 1 line ]  
^G Help      ^O Write Out ^W Where Is  ^K Cut      ^T Execute  ^C Location  
^X Exit      ^R Read File ^\ Replace   ^U Paste    ^J Justify  ^_ Go To Line
```

```
sayedmohamed@sayedmohamed-Precision-3510: ~  
sayedmohamed@sayedmohamed-Precision-3510:~$ sudo nano /etc/hostname  
sayedmohamed@sayedmohamed-Precision-3510:~$
```

===== Part 2 =====

1- Change your default SELinux mode to permissive and reboot.

```
sayedmohamed@sayedmohamed-Precision-3510: ~  
sayedmohamed@sayedmohamed-Precision-3510:~$ sudo selinux-config-permissive  
[sudo] password for sayedmohamed:  
sudo: selinux-config-permissive: command not found  
sayedmohamed@sayedmohamed-Precision-3510:~$ sestatus  
SELinux status: disabled  
sayedmohamed@sayedmohamed-Precision-3510:~$
```

sudo vi /etc/selinux/config

SELINUX=permissive

```
sudo reboot
```

2- After reboot, verify the system is in permissive mode.

```
Sestatus
```

```
Current mode: permissive
```

3- Change the default SELinux mode to enforcing.

```
sudo vi /etc/selinux/config
```

```
SELINUX=enforcing
```

4- Change the current SELinux mode to enforcing.

```
sudo setenforce 1
```

```
Getenforce
```

5- Copy /etc/resolv.conf file to root's home directory.

```
sudo cp /etc/resolv.conf /root/
```

6- Observe the SELinux context of the initial /etc/resolv.conf

```
ls -Z /etc/resolv.conf
```

7- Move resolv.conf from root's home directory to /etc/resolv.conf

```
sudo mv /root/resolv.conf /etc/resolv.conf
```

8- Observe the SELinux of the newly copied /etc/resolv.conf

```
ls -Z /etc/resolv.conf
```

9- Restore the SELinux context of the newly positioned /etc/resolv.conf

```
sudo restorecon -v /etc/resolv.conf
```

10- Observe the SELinux context of the restored /etc/resolv.conf

```
ls -Z /etc/resolv.conf
```

11- Configure OpenSSH to allow public key-based login credentials

```
sudo vi /etc/ssh/sshd_config
```

```
PubkeyAuthentication yes
```

```
AuthorizedKeysFile .ssh/authorized_keys
```

```
sudo systemctl restart sshd
```

12- Create an SSH key-pair

```
ssh-keygen -t rsa -b 4096 -f ~/.ssh/id_rsa
```

13- Configure to login without the need of a password.

```
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
```

```
chmod 600 ~/.ssh/authorized_keys
```

```
chmod 700 ~/.ssh
```

14- Configure SSH to prevent root logins.

```
sudo vi /etc/ssh/sshd_config
```

```
PermitRootLogin no
```

```
sudo systemctl restart sshd
```